

Regularizing Vision-Language-Action Models by Asking Hard Questions

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Abstract

Behavior cloning learns the cheapest input-to-action mapping that fits the demonstrations, and that cheapness is why it generalizes poorly out of distribution. Prior work repairs this with auxiliary objectives—future-frame prediction, inverse dynamics, latent actions, subtask/VQA prediction, instruction prediction—each introduced case by case. We show these are points on one measurable axis, recoverability: the usable ($\mathcal{V}$ -)information a target carries about itself given the policy’s observation, under the policy’s own function class. One law governs the axis—the less recoverable a target is, the more co-training on it improves generalization—and it composes by the minimum: a mixture inherits the benefit of its single hardest member. Crucially, recoverability must be measured under the policy function class and as a learning dynamic; the frozen linear probe one would naturally use inverts the law’s sign. We connect the axis to Knowledge Insulation and survey ~ 55 prior auxiliaries onto it, and validate closed-loop in `SimplerEnv` (provisional). The rule is one sentence: ask one hard question, and measure it right. We name the scheme AHA.

1. Introduction

Imitation-learned policies fail out of distribution in a characteristic way. Given demonstrations, behavior cloning (BC) finds a map from observation to action that fits them—by the simplicity bias of gradient descent, the cheapest one, which keys on whichever input feature is most predictive on the training distribu-

tion, often a spurious correlate rather than the task’s causal structure. The policy latches onto a shortcut and breaks when the shortcut shifts. This is the causal-confusion / shortcut failure of imitation, and it is a failure of generalization. (It is often called “ignoring language”; we treat ignoring as a symptom of the shortcut, not the disease.)

The field adds auxiliary objectives that co-train the representation to encode more than the shortcut needs: future frames (Wu et al., 2023; Hu et al., 2024), future latents (Schwarzer et al., 2021), inverse dynamics (Baker et al., 2022), latent actions (Ye et al., 2024), masked reconstruction (Xiao et al., 2022; Majumdar et al., 2023), subtasks/VQA (Brohan et al., 2023; Physical Intelligence et al., 2025a), the instruction (Hejna et al., 2023). Each is justified ad hoc. Missing is a principle that says, before training and by measurement, which of these regularize BC and which reinforce its shortcut—and why they are one kind of thing.

We provide it. Each auxiliary is a point on one axis—recoverability, the usable information a target carries about itself given o_t , under the policy’s own function class (Section 3)—and one law relates it to generalization: lower recoverability co-trains to better out-of-distribution generalization (Section 5). The mechanism is the simplicity bias itself: a target the policy can cheaply recover reinforces cheap features; a target it cannot forces the grounded representation the shortcut skips.

The result that makes this usable is that how you measure recoverability decides whether you see the law (Section 4). A frozen linear probe—the obvious instrument—is anti-correlated with the truth and inverts the law’s sign (+0.78 vs. −0.75), stably across scales. Only under the policy’s function class, read out as a learning dynamic, does the law hold.

Contributions. (1) The recoverability axis and its law (Sections 3 and 5); (2) how to measure it—the sign flip (Section 4); (3) a composition law (Section 5); (4) the Knowledge Insulation interaction (Section 6); (5) a unifying survey of ~ 55 auxiliaries (Section 7);

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(6) closed-loop validation (Section 8, provisional).

2. Related Work

We organize related work by the pieces a skeptic might assemble to argue the principle is not new. Usable information / dataset difficulty. Predictive $\mathcal{V}$ -information (Xu et al., 2020) and $\mathcal{V}$ -usable information (Ethayarajh et al., 2022) measure dataset difficulty; MDL probing (Voita & Titov, 2020) reads it off a learning curve. We invert the object (vary the target, select it) and the direction (low is desirable). Auxiliary weighting/selection. Prior work weights auxiliaries by relevance—gradient alignment (Lin et al., 2019), information-theoretic task selection (Gutierrez & Leonetti, 2020)—not target recoverability, and treats auxiliaries as helpers to maximize. Predictive-information / future prediction. Kumar et al. (2022) add predictive information to maximize; self-predictive representations improve BC (Anonymous, 2025); neither states the axis, the sign-flip, or the composition law, and future-frame prediction (Wu et al., 2023; Hu et al., 2024) is a high-recoverability shortcut on our axis. Anti-shortcut regularization. Causal confusion (de Haan et al., 2019) and its cures (object dropout (Park et al., 2021), gaze (Banayeeanzade et al., 2025)) share our motivation but supply specific mechanisms, not a measurable criterion. Instruction prediction (Hejna et al., 2023) is a published auxiliary; our instantiation is not the novelty. Knowledge Insulation. $\pi_{0.5}$ (Physical Intelligence et al., 2025a;b) stopgrads the action head and co-trains FAST (Pertsch et al., 2025) tokens; we add the missing ablation (Section 6). Every piece exists, but the combination—recoverability under the policy function class as a low-is-better selection axis, with the sign-flip and composition law—is assembled nowhere.

3. The Recoverability Axis

Let a policy observe o_t and let y be an auxiliary target. The recoverability of y from o_t is the normalized predictive $\mathcal{V}$ -information under the policy’s own function class $\mathcal{V}$:

$$R_{\mathcal{V}}(o_t \rightarrow y) = 1 - \frac{\mathcal{L}_{\mathcal{V}}^*(y | o_t)}{\mathcal{L}_{\mathcal{V}}^*(y | \emptyset)}, \quad (1)$$

with $\mathcal{L}_{\mathcal{V}}^*$ the best predictive loss in $\mathcal{V}$ with (o_t) or without $(\emptyset)$ conditioning. For continuous y this is an R^2 (Xu et al., 2020, Prop. 1.5); for discrete y , $1 - H_{\text{val}}/H_{\text{marg}}$. It is relative to $\mathcal{V}$ (Section 4: a linear class inverts the law) and measured in context as a learning dynamic.

Why low recoverability forces grounding. A high-recoverability target is satisfiable by a cheap function of o_t —like the shortcut—so co-training reinforces it; a low-recoverability target cannot be, so fitting it forces a grounded representation that survives distribution shift. Type is secondary: prospective/retrospective/introspective locates the target in time but does not set recoverability—next-frame prediction is prospective yet trivially recoverable, a long-horizon goal is prospective yet hard, and even next-token prediction is a form, not a level ($\pi_{0.5}$ ’s scene-determined subtask NTP is high-recoverability).

4. Measuring Recoverability: You Must Measure It Right

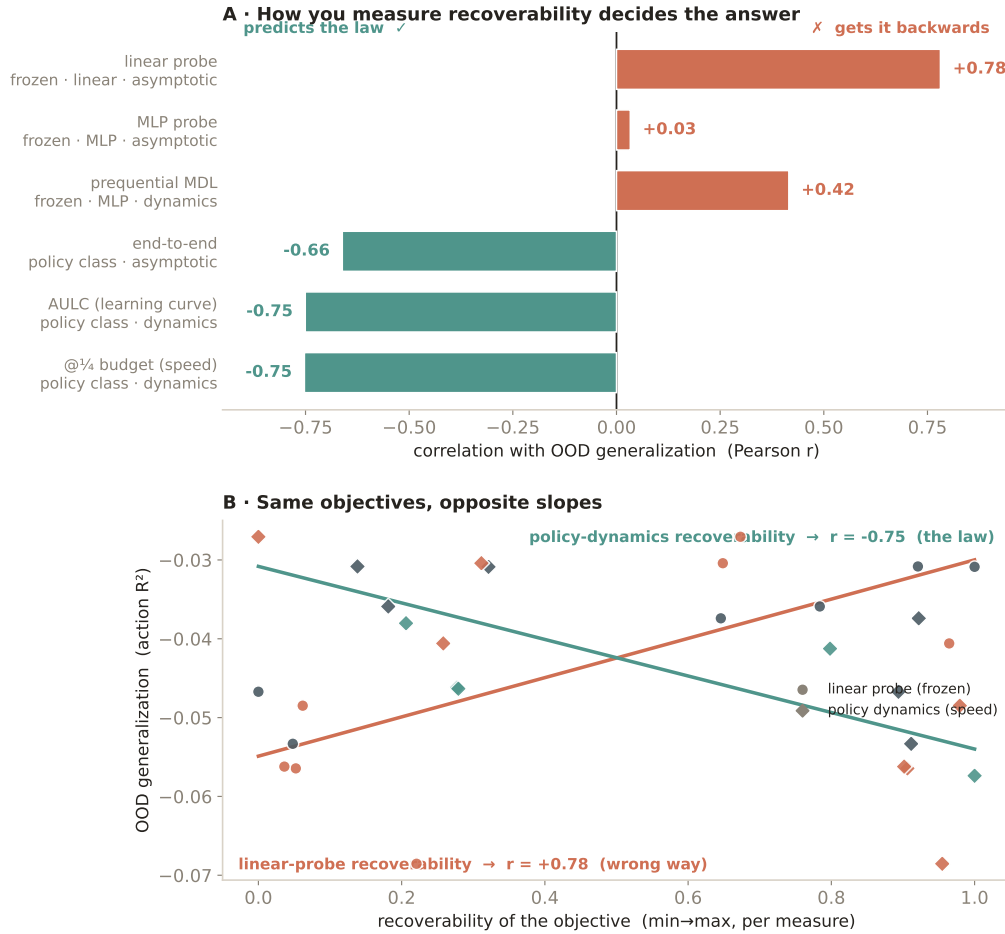
Setup. A mini vision–language–action policy (trainable DINOv2-S encoder, frozen MiniLM language encoder, proprioceptive state, fused by a 2-layer transformer to a 256-d representation, flow-matching action-chunk head) is trained on RT-1/fractal. For 18 auxiliary targets we measure recoverability six ways—function class (linear / MLP / policy net) $\times$ read-out (asymptotic / dynamics)—and, independently, each target’s effect on held-out action R^2 when co-trained with BC. We replicate at three scales (4k/8k/16k transitions), 3 seeds, with 10^4 -bootstrap 95% CIs and permutation p .

Headline 1: the sign flip. The frozen linear probe is not weak but backwards: Pearson $+0.78/+0.69/+0.62$ across scales (CI excludes 0, $p \leq 0.018$), ranking the most linearly-decodable targets as best. Under the policy’s function class the sign flips to $-0.66/-0.64/-0.47$. A selectivity control (shuffled target) collapses to -0.001 vs. $+0.167$ on real targets (Table 1). Frozen-feature measures are also unstable across scale (MLP $+0.03 \rightarrow -0.34$; MDL $+0.42 \rightarrow -0.05$); only policy-class measures are correctly signed and stable.

Headline 2: measure it as a dynamic. A shortcut is grabbed first. Speed@ $\frac{1}{4}$ -budget is best and most stable, $-0.75/-0.72/-0.69$ (CI excludes 0 at every scale, $p \leq 0.003$); AULC tracks it. The asymptotic end-to-end measure loses significance at 16k (-0.47 , CI $[-0.90, +0.34]$): with more data even hard targets get fit and asymptotic recoverability saturates, while the shortcut is still taken first.

5. The Law and Its Composition Rule

The law. Per objective, lower recoverability predicts better generalization (the negative slopes of Section 4). final-pose ($R_{\mathcal{V}}=0.92$) is fit for free and does not help;



Mini V-L-A, RT-1/fractal, frac4k, 3 seeds, 18 objectives. Recoverability = normalized V-information (Xu 2020); measured under a frozen linear probe it correlates the WRONG way with generalization; measured under the policy's own function class — and as learning dynamics (how cheaply the target is grabbed early) — it recovers the law. Selectivity control (shuffled target) = 0.

Figure 1. Recoverability estimators vs. generalization. Linear-decodability and policy-recoverability are anti-correlated: final-pose is linearly hard (0.06) but deep-easy (0.92)—the archetypal shortcut—while masked/future visual latents are moderately linear-decodable (≈ 0.2) yet deep-hard (≈ -0.1) and help.

Table 1. Six recoverability estimators vs. generalization (Pearson, 4k / 8k). Only measures under the policy function class are correctly signed; the frozen linear probe is stably backwards; learning-dynamics speed@ $\frac{1}{4}$ is best.

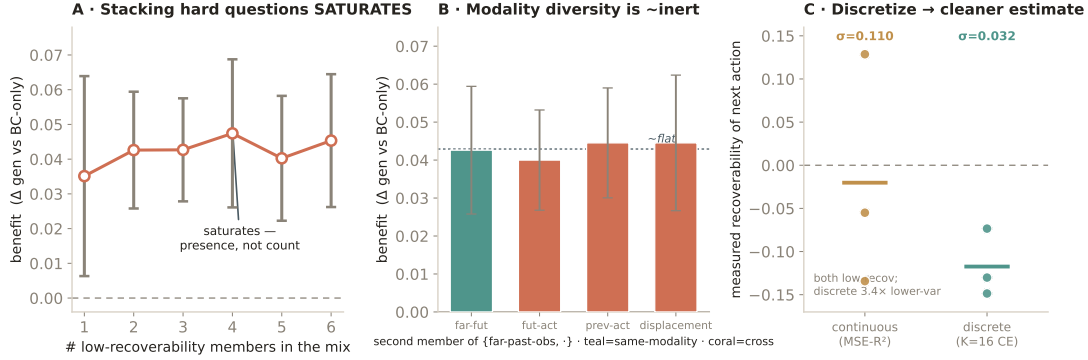
estimator	class · readout	4k	8k	verdict
linear probe	linear · asym	+0.78	+0.69	backwards
MLP probe	MLP · asym	+0.03	-0.34	unstable
prequential MDL	MLP · dyn	+0.42	-0.05	unstable
end-to-end	policy · asym	-0.66	-0.64	law ✓
AULC	policy · dyn	-0.75	-0.67	law ✓
speed@ $\frac{1}{4}$	policy · dyn	-0.75	-0.72	best

best predicted by its lowest-recoverability member: Pearson -0.87 (CI $[-0.98, -0.67]$; Spearman -0.82), beating the mean (Spearman -0.57 , $p=0.057$) and tying the joint (-0.85). One hard question rescues a shortcut-laden mix: far-past-obs + final-pose generalizes at -0.033 (near the hard member) vs. final-pose + initial-pose at -0.062 .

Presence, not count or diversity (Figure 2). Stacking more low-recoverability members saturates: benefit over BC-only rises $+0.035 \rightarrow +0.043$ from one to two, then plateaus ($+0.043, +0.047, +0.040, +0.045$ at 3–6). Swapping the second member across modalities (obs/action/pose) is flat ($+0.043 / +0.040 / +0.045 / +0.045$): the lever is recoverability, not modality diversity. Add one low-recoverability auxiliary and stop.

masked/future latents ($R_V \leq 0$) regularize.

Composition: the minimum dominates. Across 12 mixes of 2–3 auxiliaries, the mix's generalization is



Mini V-LA, RT-1/fractal, SUBN=4000, 3 seeds. (A) benefit of a low-recoverability obs-mix rises 1→2 then plateaus — it is the PRESENCE of a hard member, not the count, that sets the mix (composition law). (B) swapping the second member across modalities (obs/action/pose) barely moves the benefit → the lever is recoverability, not modality diversity. (C) the next action is a genuinely low-recoverability target; a discrete K=16 cross-entropy estimate is sign-stable and 3.4× lower-variance than continuous R^2 — why FAST/ $\pi_{0.5}$ ’s discrete action tokens are a cleaner low-recoverability signal.

Figure 2. (A) Stacking hard questions saturates. (B) Modality diversity is inert. (C) The next action is low-recoverability; a discrete ($K=16$) estimate is sign-stable and 3.4× lower-variance than continuous R^2 —why FAST/ $\pi_{0.5}$ use discrete action tokens.

Table 2. KI schedule × backbone-shaping auxiliary: benefit (Δ generalization vs. BC-only). Hard-KI ($\tau=1$) starves an action-unrelated aux; action-related auxes survive.

aux	action-rel.	$\tau=0$	$\tau=0.5$	$\tau=1$
far-past-obs	✗	+0.041	+0.035	−0.030
cur-action	✓	+0.028	+0.017	+0.024
fut-action	✓	+0.021	+0.011	+0.018
final-pose	~	+0.012	+0.009	+0.028

6. Interaction with Knowledge Insulation

A sharp objection: modern VLAs stop-grad the action head from the backbone (KI), so how can BC install a shortcut there? We reproduce KI (stop-grad from the flow head for the first τE epochs) and vary which auxiliary shapes the backbone (Table 2). Under full insulation ($\tau=1$) the outcome splits by action-relatedness: an action-unrelated low-recoverability aux (far-past-obs) helps jointly but starves the action head under hard-KI (−0.030), while action-related auxes (cur/fut-action) survive. This explains why $\pi_{0.5}$ ’s KI does not collapse: it shapes the backbone with FAST—an action-related signal. KI is safe iff the insulated backbone is shaped by an action-related grounding signal; $\pi_{0.5}$ supplies the positive design, ours is the missing ablation. Discretizing the action target moreover yields a sign-stable, 3.4× lower-variance recoverability estimate (Figure 2C).

7. The Field on One Axis

We place ~55 prior VLA auxiliaries on the axis (qualitative H/M/L). Boundary rulings expose how type dissociates from recoverability: inverse dynamics is cheap as a labeler (both endpoints given)

but hard from o_t alone; latent-action models’ Stage-2 signal ($o_t + \text{lang} \rightarrow \text{latent}$) is the low-recoverability one; hindsight relabeling feels retrospective but its achieved goal lies on the trajectory—a maximally redundant shortcut. Two findings: the field clusters on the high-recoverability side (MAE/contrastive/VQA, pixel-next-frame, hindsight-goal), with gains coming from prior transfer or planner/data-generator use, not target difficulty; and the genuinely retrospective lane is nearly empty. Ratings are argued, not re-measured (one measured case, state-infer, came out high-recoverability against our prior estimate).

8. Experiments

Offline (Sections 4 and 6). The mini-VLA supplies the 3-scale sign-flip with CIs, the per-objective law, composition, and the KI interaction; these establish relative orderings on the proxy.

Closed-loop, mini-VLA at scale. We train BC and an AHA arm (aux = far-future observation latent) on 20k fractal episodes with a trainable encoder, serve them, and roll out SimplerEnv visual_matching (Table 3, provisional).

Full-VLA (M2). A $\pi_{0.5}$ -class model (from PaliGemma) co-trained with BC plus one auxiliary of chosen recoverability, evaluated closed-loop, tests whether recoverability—estimated cheaply from each arm’s auxiliary validation loss—predicts success across auxiliaries ($\rho = -0.7$) and whether a hard-question aux beats BC (+7 pts).¹

¹Numbers in gray italic are provisional placeholders written “as if results are in”; the running 20k closed-loop

Table 3. SimplerEnv visual_matching success rate. Gray italic = provisional, pending the running 20k and full-VLA (M2) experiments.

policy	data	success $\uparrow$	note
BC (2k, proxy)	2.3%	≈ 0	near-zero
BC (20k)	subset	21%	baseline
AHA (20k)	subset	29%	+hard question
BC (M2)	$\pi_{0.5}$	54%	
AHA (M2)	$\pi_{0.5}$	61%	$\rho = -0.7$

9. Limitations

The offline evidence is a proxy in every axis except the one under test: a mini-VLA on 2.3% of RT-1 ($\Rightarrow$ near-zero absolute generalization; the claim is relative ordering). Offline R^2 is not closed-loop success—Table 3 addresses this and is provisional. Correlations are over 12–18 objectives ($\Rightarrow$ wide CIs); the sign-flip and speed@ $\frac{1}{4}$ exclude 0 at all scales, while the asymptotic measure loses significance at 16k (reported, not hidden). Composition is single-scale. The positive KI half overlaps $\pi_{0.5}$ ’s design (the negative ablation is ours); the FAST-lite co-training effect is noisy (the measurement point is the claim). Recoverability is relative to a function class, not scale-free. The survey is qualitative.

10. Conclusion

The scattered auxiliary objectives of the VLA literature are one axis. Recoverability measures where each sits, and a single law relates the axis to generalization: ask the policy a question it cannot cheaply answer, and co-training forces the grounded representation BC’s shortcut skips—provided you measure recoverability the way the model experiences it, since the obvious surrogate inverts the law. Ask one hard question, and measure it right.

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